SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0706

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards (BL2, BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks: 25

Annexure A

Comparative Analysis

Code of Conduct: *V.Lasya(192524186)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action*

Signature of the student: v.lasya

Annexure A — Comparative Analysis

The four fundamental network topologies — star, bus, mesh, and hybrid — differ significantly in cost, reliability, performance, fault tolerance, installation complexity, scalability, and maintenance. Each comparison below is supported with a summary table and a network diagram.

1. Star Topology vs Bus Topology — Cost, Reliability & Performance

Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub, along with more cabling to connect each node individually. Bus topology, in contrast, uses a single shared backbone cable, which keeps it low-cost and simple to install. However, star topology is far more reliable: the failure of one node or its cable does not affect any other device, whereas a break anywhere in the bus backbone can bring the entire network down. Performance also favours star topology, since the central switch manages and isolates traffic, reducing collisions — while bus topology degrades noticeably as more devices are added and contend for the same shared medium.

Parameter	Star Topology	Bus Topology
Cost	Higher (switch/hub + more cable)	Lower (single backbone cable)
Reliability	High — one node's failure is isolated	Low — backbone failure disrupts all nodes
Performance	Better — central device manages traffic	Degrades as more devices share the medium

Table 1: Star vs Bus — Cost, Reliability and Performance

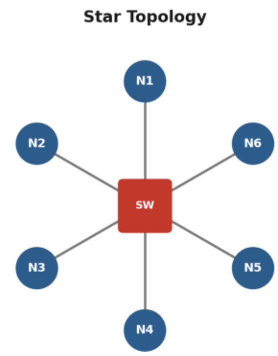

Fig. 1(a): Star Topology — all nodes connect to a central switch/hub

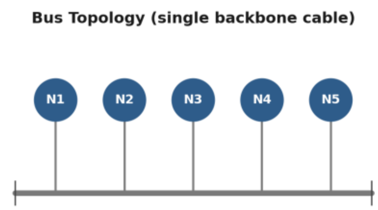

Fig. 1(b): Bus Topology — all nodes share a single backbone cable

2. Mesh Topology vs Star Topology — Fault Tolerance

Mesh topology offers very high fault tolerance because every node connects to multiple other nodes, creating several alternate paths for data. If one link fails, traffic can simply be rerouted through another path without any disruption to communication. Star topology, by comparison, has only moderate fault tolerance — an individual node's failure does not affect the rest of the network, but failure of the central hub or switch stops the entire network from functioning. This makes mesh topology considerably more robust and dependable for critical, high-availability environments.

Parameter	Mesh Topology	Star Topology
Redundant paths	Multiple paths between every pair of nodes	Only one path — through the central device
Single point of failure	None	Central hub/switch
Fault tolerance	Very high	Moderate

Table 2: Mesh vs Star — Fault Tolerance

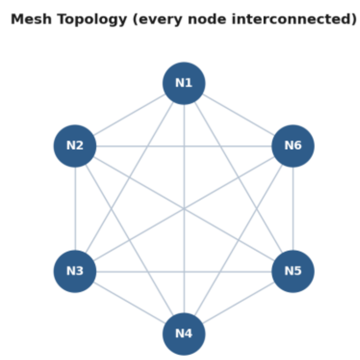

Fig. 2: Mesh Topology — every node interconnected, providing redundant paths

3. Bus Topology vs Mesh Topology — Installation Complexity

Bus topology is simple and quick to install because it needs only a single backbone cable with minimal connection points, making it well-suited to small networks. Mesh topology, on the other hand, is highly complex to set up, since every node must be individually connected to every other node — requiring a very large number of cables and ports. This complexity grows rapidly as more devices are added, making mesh topology far more time-consuming and costly to implement than the straightforward wiring of a bus network.

Parameter	Bus Topology	Mesh Topology
Cabling required	Minimal (one backbone cable)	Very high — n(n-1)/2 links for full mesh
Installation effort	Low, quick to set up	High, grows sharply with node count
Suitability	Small, simple networks	Small, high-reliability critical networks

Table 3: Bus vs Mesh — Installation Complexity

4. Hybrid Topology vs Star Topology — Scalability

Hybrid topology is highly scalable because it combines two or more topologies — such as star, mesh, or bus — allowing the network to expand flexibly to meet organizational needs, with new segments added without disturbing the existing structure. Star topology is also scalable, but only to a limited extent, since it depends on the port capacity of the central device; once that device reaches its limit, further expansion requires additional switching hardware. Hybrid topology therefore offers noticeably greater scalability than a pure star design.

Parameter	Hybrid Topology	Star Topology
Basis of design	Combines multiple topologies (e.g., star + bus)	Single central device
Expansion	Flexible — new segments added independently	Limited by central device's port capacity
Scalability	High	Moderate

Table 4: Hybrid vs Star — Scalability

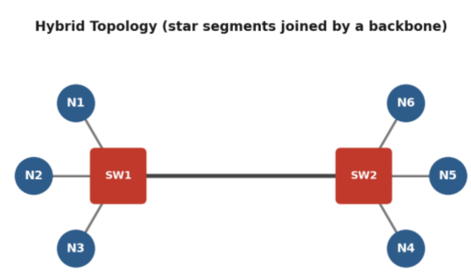

Fig. 3: Hybrid Topology — two star segments joined by a backbone link

5. Maintenance Comparison Across All Four Topologies

Star topology is relatively easy to maintain because problems can be quickly identified and isolated to a single node or cable. Mesh topology, despite its high reliability, is difficult to maintain owing to its complex structure and large number of connections, which makes troubleshooting time-consuming. Bus topology is harder to maintain than star topology because a fault anywhere in the backbone cable can disrupt the whole network and can be difficult to locate. Hybrid topology carries moderate maintenance complexity — it benefits from flexibility, but managing several combined topologies together requires careful monitoring and skilled administration.

Topology	Fault Isolation	Troubleshooting Effort	Maintenance Level
Star	Easy — isolated to node/cable	Low	Easy
Bus	Hard — backbone faults affect all	High	Difficult
Mesh	Complex — many links to check	Very high	Difficult
Hybrid	Depends on combined segments	Moderate	Moderate

Table 5: Maintenance Comparison — Star, Bus, Mesh and Hybrid

Overall Summary

The table below consolidates all comparison parameters discussed above across the four topologies for quick reference.

Criteria	Star	Bus	Mesh	Hybrid
Cost	High	Low	Very High	High
Reliability	High	Low	Very High	High
Performance	Good	Degrades with load	Excellent	Good
Fault Tolerance	Moderate	Low	Very High	High

Criteria	Star	Bus	Mesh	Hybrid
Installation Complexity	Moderate	Low	Very High	High
Scalability	Moderate	Low	Low	High
Maintenance	Easy	Difficult	Difficult	Moderate

Table 6: Overall Comparison of Star, Bus, Mesh and Hybrid Topologies